Oluwatosin Ayeni

AI Inference Engineer — Fuse Energy

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PROFILE

Founding-calibre AI Inference Engineer with deep expertise in production LLM serving, autoscaling infrastructure, and performance optimisation. Built and ran complete GKE inference platform (NVIDIA L4 24 GB): vLLM 0.32.0 + SGLang 0.5.20 serving Qwen3 8B AWQ and Llama 3.1 8B AWQ; LiteLLM Proxy gateway with least-busy routing, A/B weighted round-robin, Redis response cache, and PostgreSQL spend tracking; KEDA autoscaling on queue depth; 27-target Prometheus + Grafana + AlertManager + Loki. LIVE: 723 requests, 1.25M+ tokens, $28.74/day. H100 NVL benchmarks: 5,965 tok/s peak, 1,443× TTFT speedup via RadixAttention, 100% SLO compliance at concurrency 128. Experienced owning the full serving stack — from hardware utilisation and KV cache tuning to routing policies, SLA enforcement, and capacity planning.

CORE COMPETENCIES

Inference Serving	vLLM, SGLang, TensorRT-LLM, LiteLLM Proxy — production deployment, benchmarking, routing, SLA management
Serving Optimisation	KV cache management, continuous batching, quantisation (AWQ/GPTQ), speculative decoding, prefix caching
Autoscaling & Capacity	KEDA autoscaling on queue depth, HPA on GPU node pools, Kubernetes, capacity planning, cost-efficiency tuning
GPU & Profiling	NVIDIA Nsight Systems/Compute, CUDA kernel profiling, VRAM utilisation, BFloat16/FP8, DCGM Exporter
Observability & Ops	Prometheus, Grafana, AlertManager, Loki, FluentBit, per-node weight caching, Argo CD GitOps, Terraform

PROFESSIONAL EXPERIENCE

Jan 2025–present — **Founder & Lead Engineer** — *Nexora Academic*, Glasgow, UK

- H100 NVL benchmark: vLLM v0.30 vs SGLang v0.5.20 — Qwen3-8B BF16, TP=2. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance up to concurrency 128. Live: inference-web-theta.vercel.app
- Built *inference-optimizer*: production AI Inference Gateway — 3.87× faster (parallel execution), 61.4% cost reduction (prompt caching), 14× throughput (concurrency), 64.3% cheaper (model routing). 13 optimisations, 4 providers.
- Built *vllm-k8s-stack*: Kubernetes LLM serving — KV-cache-aware routing, HPA autoscaling on GPU pools, Prometheus/Grafana observability.
- Built *inference-harness*: unified benchmark framework spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard.
- Designed 13-phase AI Inference Engineering curriculum covering GPU architecture, CUDA optimisation, serving frameworks, and distributed inference.

Jan 2023–present — **Founder & AI Consultant** — *Vertinab / AnaburyAI Technology*, Glasgow, UK

- 28× LLM cost reduction deploying Qwen2.5-7B on RTX 4090 via vLLM/SGLang with continuous batching and KV cache optimisation.
- CUDA KV cache kernel profiling with NVIDIA Nsight: identified memory bandwidth bottleneck; 3.8× throughput improvement; 18× latency reduction via continuous batching redesign.
- Fine-tuned Mistral 7B with LoRA/PEFT on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; deployed as production clinical inference API.

2024–present — **Senior Software Engineer (Contract)** — *EduTechConnect / Stellar Labs*, Remote

- Real-time NLP pipeline (WhatsApp → Zoho CRM → Supabase) with sub-500 ms end-to-end latency under concurrent load.

Mar 2026–present **AI Automation Engineer** — *EduTech Global*, Remote

- LLM-backed student chatbots with structured output extraction and multi-step reasoning for Ahmadu Bello and Babcock universities.

2018–2022 **Data Analyst** — *Bytelab Technologies Ltd*, Nigeria

KEY PROJECTS

llm-inference-platform (GKE) Production LLM serving on GKE: vLLM 0.32.0 + SGLang 0.5.20, KEDA autoscaling, LiteLLM Proxy gateway, 27-target Prometheus stack. 723 requests, 1.25M+ tokens, $28.74/day. GitHub

llm-inference-benchmark (H100) 2× H100 NVL 94 GB, CUDA 13.4. vLLM vs SGLang, Qwen3-8B BF16, TP=2. 1,443× TTFT speedup. SLO 100% at concurrency 128. Live dashboard

vllm-k8s-stack Production K8s LLM serving: vLLM, KV-cache-aware routing, HPA autoscaling, GPU node management. GitHub

open-model-engine GPU inference benchmarking: vLLM, SGLang, TensorRT-LLM; CUDA profiling, quantisation, speculative decoding. GitHub

inference-optimizer 13-optimisation Inference Gateway: 3.87× faster, 14× throughput, 61.4% cost reduction. GitHub

EDUCATION

2022–2023 **MSc Artificial Intelligence and Applications, Distinction** — *University of Strathclyde*, Glasgow, UK

2010–2016 **MTech / BSc Physics** — *Federal University of Technology Akure*, Nigeria

OPEN SOURCE

GitHub 43 public repos — github.com/AyeniOluwatosinOlawale